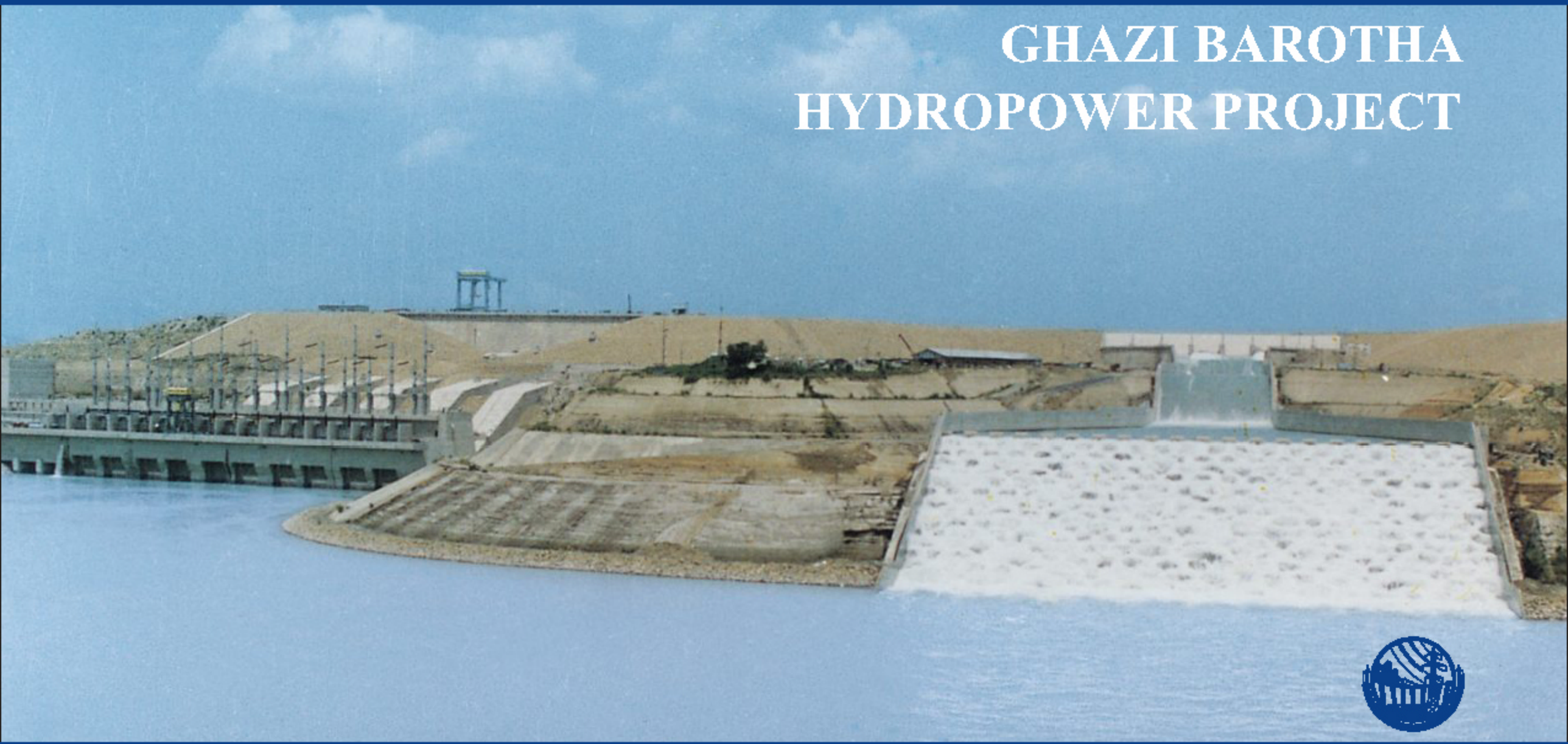
GHAZI BAROTHA HYDROPOWER PROJECT



Pakistan Water and Power Development Authority

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GBHP WAPDA, Hattian
Attoch, Pakistan

SMS Consultant 042 583 3617

Ghazi Barrage

GHAZI BAROTHA HYDROPOWER PROJECT

Introduction

The shortage of electrical power at affordable cost has long been identified as one of the main hurdles to the industrial and economic growth of Pakistan. The demand for electricity is growing rapidly and requires a considerable increase in the rate at which new generating capacity is introduced. Presently demand is met through a mix of thermal and hydroelectric plants. The percentage of thermal power generation has continued to increase in recent years, with a noticeable impact on unit cost of generation. WAPDA has continually sought to maximize the country's capacity for Hydropower generation and reduce the dependence upon thermal power generation.

Ghazi Barotha Hydropower Project with a generation capacity of 1450 MW and an average energy output of 6600 GWh is a large, renewable and emission free source of energy towards WAPDA's Vision 2025 goals.

The Project

Ghazi Barotha Hydropower Project is located on the Indus river downstream of Tarbela Dam. The Project utilizes the hydraulic head available between the tailrace at Tarbela Dam and the confluence of the Indus and Haro Rivers for power generation. In this reach Indus River drops by 76 m in a distance of 63 km. This Project possesses the minimum of environmental and social impacts.

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Head Regulator of Power Channel

The Power Channel has a nearly contour alignment with hills on the left side and the land naturally draining towards the Indus River on the right side. The Power Channel intercepts fifty three nullahs (natural drains) of which twenty-seven major nullahs have been passed over the Power Channel by providing superpassages. The remaining twenty four minor nullahs are being discharged into the Power Channel through individual inlets whilst one nullah is passing underneath the channel through a culvert.

In addition to the thirty four road bridges, including bridges for both Islamabad-Peshawar Motorway and the G.T road, there are 12 pedestrian crossings over the Power Channel.

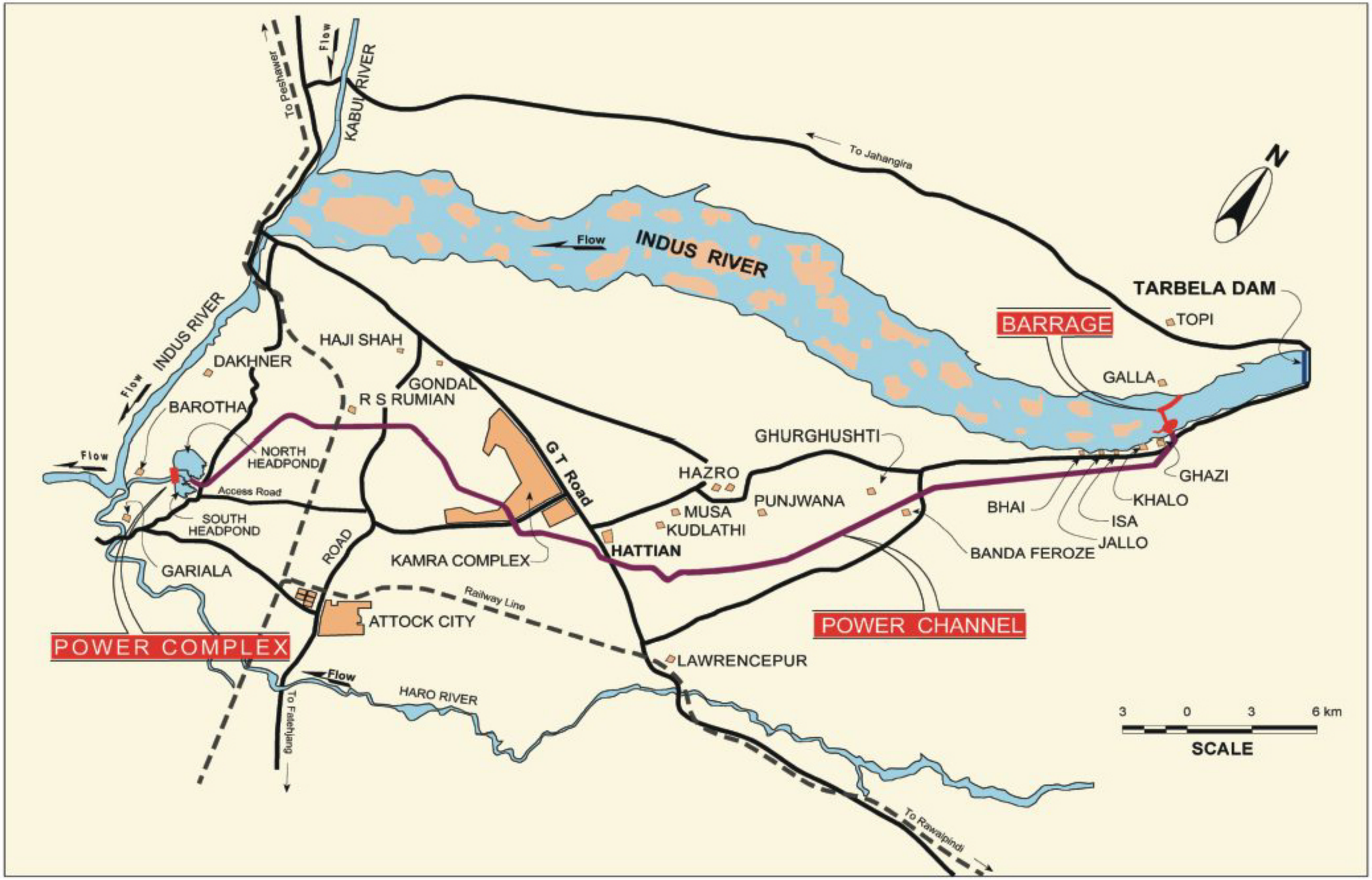
The main railway line joining Rawalpindi to Peshawar also crosses the power channel and required the construction of the second longest single span railway bridge in Pakistan. This may be the last riveted bridge of its type, constructed in Pakistan.

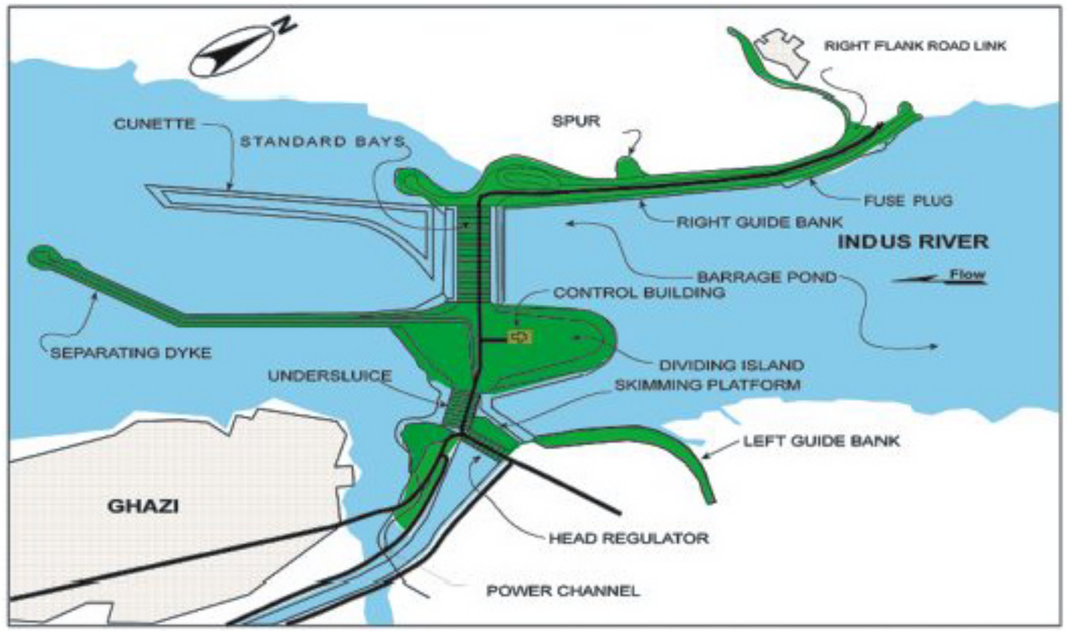
The Power Complex

The Power Complex having two headponds with a combined live storage capacity of approximately 25.5 million cubic meter is sufficient for the daily requirement of 4 hours peak generation. This means that in May and June when there is reduced generation from Tarbela and Mangla power houses, due to low reservoir levels, Ghazi Barotha Hydropower Project provides peak production of 1450 MW.

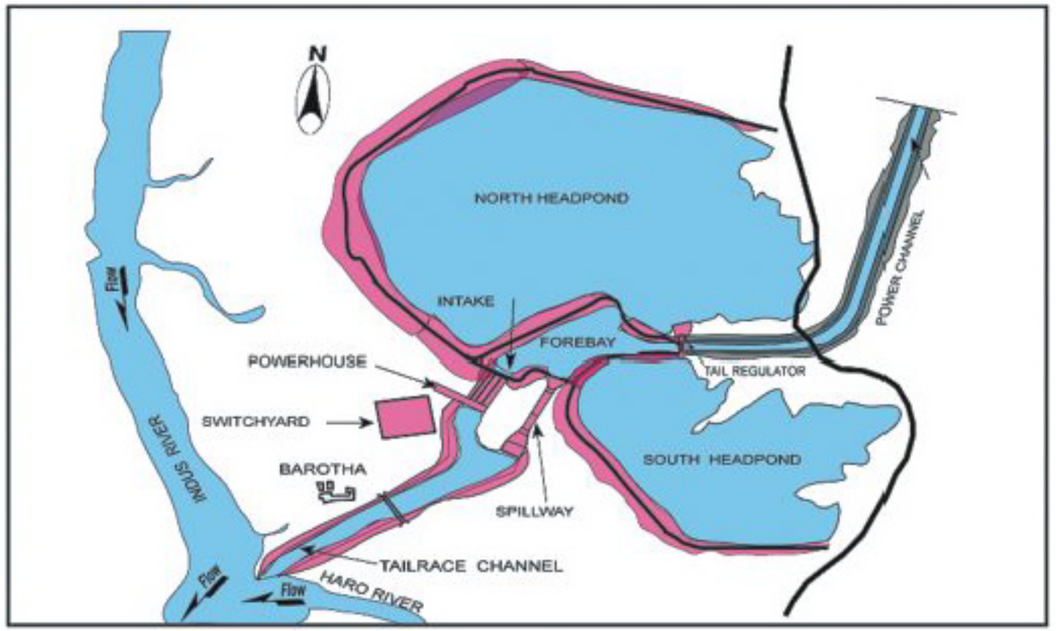
The five generating units in the Powerhouse are each fed by a 10.6 m diameter steel lined penstock. Each of the five 290 MW Turbo Generators can take a peak flow of 460 cumecs.

View of Power Channel

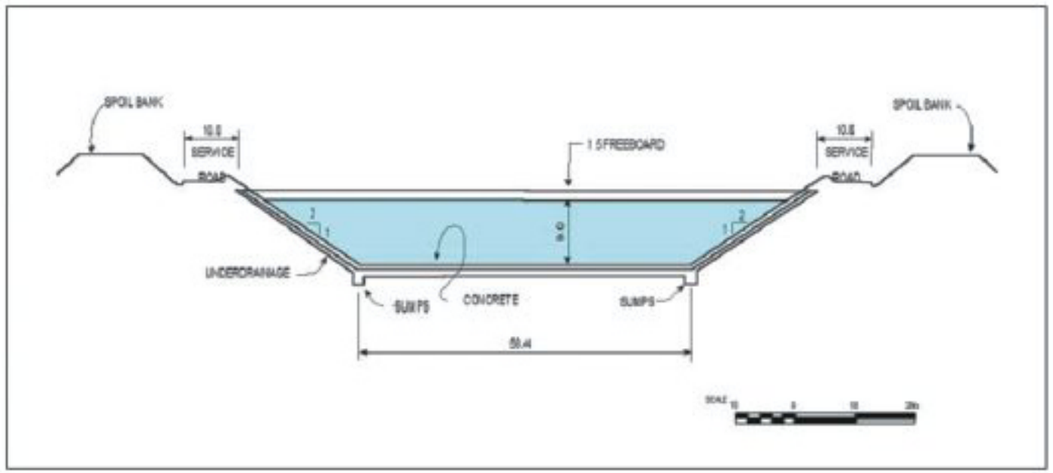

To Peshawer
Flow
KABUL RIVER
To Jahangira
N
Flow
INDUS RIVER
TARBELA DAM
BARRAGE
TOPI
GALLA
Flow
INDUS RIVER
HAJI SHAH
DAKHNER
GONDAL
R S RUMIAN
BAROTHA
GHURGHUSHTI
Flow
NORTH HEADPOND
Access Road
HAZRO
GHAZI
PUNJWANA
BHAI
KHALO
MUSA
ISA
KUDLATHI
JALLO
SOUTH HEADPOND
ROAD
GT Road
HATTIAN
BANDA FEROZE
GARIALA
KAMRA COMPLEX
Railway Line
ATTOCK CITY
POWER CHANNEL
POWER COMPLEX
LAWRENCEPUR
Flow
HARO RIVER
To Fatehjang
To Rawalpindi
3 0 3 6 km
SCALE



Barrage Layout



Power Complex Layout



Power Channel Cross Section

MISCELLANEOUS INFORMATION

EXECUTING AGENCY

Water and Power Development Authority
WAPDA House, Lahore, Pakistan

ENGINEER

Pakistan Hydro Consultants

A joint venture of
National Engineering Services Pakistan (Pvt) Ltd, Pakistan
Associated Consulting Engineers (Pvt) Ltd., Pakistan
Binnie & Partners (Overseas) Ltd, UK
Harza Engineering Company Intern'l, USA
Ewbank Preece Ltd, UK

CONSTRUCTORS

Contract C-01 Barrage and Ancillary Works and Contract C-02 Power Channel and Ancillary Works

Ghazi Barotha Contractors

A joint venture of
Impregilo, Italy
ED. Zublin AG, Germany
Nazir & Co (Pvt) Ltd, Pakistan
Saadullah Khan & Brothers (SKB), Pakistan

Contract C-03, Power Complex and Ancillary Works
Dongfang Electric Corporation, China

Contract C-03A, Penstocks
VA Tech, Austria

Contract ME-01, Turbines and Auxiliaries
Voith Hydro Germany

Contract ME-02, Generators and Auxiliaries
Mitsui/Toshiba, Japan

Contract ME-03, Transformers and Miscellaneous Equipment
Dongfang Electric Corporation, China

Contract ME-04, Cranes
Powerhouse Main Cranes
Waagner Biro AG, Austria
Powerhouse Auxiliary Cranes
Heavy Mechanical Complex, Pakistan
Gantry Cranes
CAMC, China
Trashrack Cleaning Machine
OMG, Italy
Mobile Crane
Mitsui, Japan
Truck Trailer
Volvo, Pakistan

Contract ME-05, HV Switchgear & Protection Equipment
ABB, Germany

Contract ME-08, SCADA & Telecommunication
Alstom Power Generation, Germany

Contract ME-09, Gate Equipment
CMEC, China

Contract ME-10, Transmission Line
ICC, Pakistan
Nazir & Co. Pakistan
MR Electric, Pakistan
Greaves-Potential, Pakistan

PRINCIPAL DATA

BARRAGE

Normal pond level	340.0 m
Design flood	18,700 cumecs
Survival flood	46,200 cumecs
Embankments	
Fuseplug, Rim and Dividing Island	
Standard Bays	
No. of openings	20
Crest level	332.2 m
Head Regulator	
No. of openings	8
Crest level	333.0 m
Undersluices	
No. of openings	8
Crest level	326.0 m

POWER CHANNEL

Design flow	1,600 cumecs
Longitudinal slope	1:9600
Length	51,906 m
Full supply depth	9 m
Bed width	58.4 m
Side slope	1V : 2H
Lining thickness	135 mm
Structures	
Road bridges	34
Railway bridge	1
Pedestrian Crossings	12
Superpassages	27
Drainage Inlets	24
Escapes	5
Culverts	1

POWER COMPLEX

Headponds and Forebay	
Normal pond level	334 m
Live volume	25.5 Mm^3
Siphon spillway	
Discharge capacity	1600 cumecs
Crest level	334.5 m
Intake	
No. of openings	10
Crest level	305.5 m
South Sill	
No. of openings	3
Sill level	323 m
North Sill	
No. of openings	10
Sill level	321 m

HYDROMECHANICAL EQUIPMENT

Penstocks	
No.	5
Diameter	10.6 m
Length	222 m
Turbine	
Type	Francis
Number	5
Full Gate output	295 MW
Generators / Transformers	
Number	5
Continuous Rated output	3x107.5 MVA
Rated Voltage HV:LV	515v 3:18 KV
Switchyard	
No. of 500 KV Bays	6
Scheme of Layout	Breaker & Half Scheme

units continued and Unit No. 5 was commissioned in April 2004. The construction of the North Headpond was completed in December 2004, thus completing the implementation of the Project works.

Project Financing

The PC-I of the Project was approved by Government of Pakistan in July 1994 at a total cost of Rs. 89840 million. The total cost which has been incurred on the Project is Rs. 96957 million.

The project has been funded by WAPDA supported by the following international lending agencies:

- World Bank
- Asian Development Bank
- Japanese Bank for International Cooperation
- Kreditanstalt fuer Wiederaufbau (KfW)
- European Investment Bank
- Islamic Development Bank

This Project is an important component of Pakistan's power system. The least cost status of the Project remains valid for the full range of sensitivity analysis performed. The Project has highly favourable economic parameters, it has an EIRR of 22.19% and FIRR of 13.76%. The economic and financial returns have shown that the Project forms a part of the least cost generation expansion plan for Pakistan.

Barotha Power Station

channel, terracing of waste land, reclamation of land along the left bank of Indus river and filling of deep gullies, whilst providing an environmentally satisfactory permanent solution. The spoil banks and gullies have been leveled, covered with topsoil and to be provided with tubewell irrigation. It has been planned to sell back these developed banks to the interested affectees for the cultivation purposes.

Against this disruption to the local community the construction works employed more than 10,000 project affected persons and many local companies. Over the course of the construction, this influx of capital into the local economy has had a significant effect on improving the business opportunities and economic growth particularly in the project area.

To avoid the environmental concerns of the people residing along the banks of Indus River downstream of Ghazi Barrage, compensation water is being released through barrage into the original course of the river. For the villages on the both sides of power channel, 46 No. of crossings have been provided over the power channel.

Project Implementation

In the Feasibility Report, the time for Project implementation of Civil and M&E contracts was estimated as 63 months starting from 01 April 1993. With the approval of PC-1 of Preparatory Works including land acquisition, relocation, resettlement, construction of colonies and other infrastructure for the Project were commenced in 1995. The implementation of the Project ultimately came to fruition with the inauguration ceremony of the Commissioning of Unit No. 1 and Unit No. 2 on 19 August 2003 by the President of Pakistan. The work on the commissioning of other

Environmental, Resettlement and Social Aspects

The guiding principle for Ghazi Barotha Hydropower Project has been to maintain close contact between the engineering and environmental planners, social scientists, the local community groups and NGOs, right from the feasibility stage to Project construction.

This process allowed the planning teams to identify and avoid or mitigate, all potentially serious, adverse environmental, social and archaeological effects. The locations for Barrage and Power Complex as well as the alignment of the Power Channel have been selected in such a way that it avoids where possible, the disruption of villages, cultural properties and other infrastructures. However, only 110 dwellings were affected, and for them three resettlement villages have been established in the vicinity of the Project area where all the basic amenities like water supply, sewerage, schools, mosques etc. have been provided by WAPDA to the affected households resulting no out of area resettlements.

A Project NGO, namely Ghazi Barotha Taraqiati Idara (GBTI), was established and funded by WAPDA to assist in mitigating the genuine public concerns on the matters relating to land valuation and compensation, displacement of affectees and resettlement, loss of livelihood, employment and other social and environmental concerns. In addition to this, GBTI has implemented an integrated regional development plan and carried out development activities in the project affected areas.

The power channel alignment is mostly in cut, having a total of 76 million cubic meter of excess spoil material extracted out of the channel excavation. To avoid hauling the material over long distances, the spoil material has been used for construction of spoil banks along the power

Power Complex has been provided with a self priming siphon spillway of 1600 cumecs capacity, having energy dissipation in a stilling basin and a baffle chute.

The power transmission is through 500 KV double circuit lines to WAPDA's national grid system.

Mechanical and Electrical Equipment

The installed power generating capacity is 1450 MW consisting of five units each of 290 MW. The units have a design flow of 400 cumecs at optimum gate opening and 460 cumecs at full gate opening for a design head of 69 m.

The principal items of power equipment are as follows:

- Five Francis turbines each with a 290 MW generator which together have a combined power generating efficiency of 94%.
- Five three-phase banks of transformers, each single-phase unit being 107.5 MVA.
- 500 KV conventional outdoor switchgear configured in one-and-a-half breaker arrangement.
- 12 cranes with lifting capacities from 6 to 450 tons.

With a project which is spread over such large area, it was considered necessary both for safety and efficiency that it should be monitored and controlled centrally. This has been achieved by providing two independent distributed control systems (DCS) one each at the Barrage and Power Complex which share information through an optic fiber cable link.

Tail Regulator water entering into Forebay

North Headpond

Railway Bridge crossing over Power Channel

